Circular Dichroism: Protein Secondary Structure

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Semester 2, AY2025/26

Circular dichroism (CD) spectroscopy was used to compare how acidification to pH 1 changes the secondary structure of BSA and lysozyme. Spectra were recorded from 200 to 250 nm on a JASCO J-810 spectropolarimeter. For BSA, the magnitude of the ellipticity minimum at 210 nm fell by (21.4 ± 0.3)% and the 222 nm α-helix band by (23.1 ± 0.3)%, indicating a substantial loss of α-helix. For lysozyme the minimum changed by only (1.0 ± 0.7)%, but the 222 nm band fell by (7.3 ± 0.7)%: its secondary structure changed much less than that of BSA, but measurably. These uncertainties include spectral noise only. The smaller response of lysozyme is consistent with stabilisation by its four disulfide bonds, although this comparison alone does not isolate their role.

1. INTRODUCTION

Circular dichroism measures the differential absorption of left- and right-circularly polarised light by optically active molecules. In proteins, the peptide bond chromophore gives rise to characteristic CD spectra that depend on the backbone dihedral angles, and therefore on the secondary structure. Each structural motif (α-helix, β-sheet, random coil) produces a distinct spectral signature in the far-UV region (190–250 nm), making CD a sensitive probe of secondary-structure content [1,2].

When a protein unfolds, its ordered secondary structure converts to a disordered random coil, and the CD spectrum changes accordingly. Comparing spectra of native and denatured protein directly reveals the extent of secondary-structure loss. This is complementary to fluorescence spectroscopy, which reports on tertiary-level packing around specific residues; CD instead reveals whether the backbone geometry itself has been disrupted.

BSA and lysozyme are again subjected to acid denaturation at pH 1. BSA is predominantly α-helical [3], so it should show a pronounced CD response to unfolding. Lysozyme has a smaller helical fraction, some β-sheet, and four disulfide bonds that covalently cross-link its chain [4]. The comparison asks whether the cross-linked protein keeps more of its secondary structure under the same acid treatment.

2. THEORY

2.1 Circularly Polarised Light and Optical Activity

Linearly polarised light can be decomposed into two equal-amplitude circularly polarised components, one rotating left (L) and one rotating right (R). In a non-optically-active medium, both components propagate identically and recombine to give linearly polarised light at the exit. In an optically active medium, the refractive indices and absorption coefficients for L and R differ. This difference in absorption is the physical basis of circular dichroism.

If A_L and A_R are the absorbances for the two components, the CD signal is:

$$\mathrm{CD} = A_L - A_R = \Delta A \tag{1}$$

Because L and R are absorbed unequally, the transmitted components have different amplitudes, and their superposition produces elliptically polarised light rather than linearly polarised light. The ellipticity θ of this ellipse is related to the differential absorbance by:

$$\theta = 33.0\,\Delta A \tag{2}$$

where θ is in degrees and ΔA is in absorbance units [5]. The instrument reports millidegrees, so $\theta_{\mathrm{mdeg}} = 3.30 \times 10^4\,\Delta A$. From Beer–Lambert's law, $\Delta A = \Delta\varepsilon\, c\, l$, where $\Delta\varepsilon$ is the difference in molar extinction coefficients for L and R light, c is the molar concentration, and l is the path length.

Eq. (2) is the central relation connecting the measured quantity (ellipticity) to the molecular property (differential extinction). Since secondary-structure elements impose specific constraints on the peptide-bond geometry, $\Delta\varepsilon$ varies systematically with backbone conformation, and the CD spectrum therefore encodes secondary-structure content.

2.2 Secondary-Structure Signatures

The far-UV CD spectrum of a protein arises primarily from the $n \to \pi^*$ and $\pi \to \pi^*$ transitions of the amide bond. Each secondary-structure type produces a characteristic pattern [2]. An α-helix gives strong negative bands near 208 nm and 222 nm and a positive band near 193 nm; the 222 nm band comes from the $n \to \pi^*$ transition, while the 208 nm band reflects exciton splitting of the $\pi \to \pi^*$ transition in the helical array of peptide bonds. A β-sheet gives a weaker negative band near 218 nm and a positive band near 195 nm, and a random coil a strong negative band near 198 nm with weak positive signal above 210 nm.

When a protein unfolds from a structured to a disordered state, the CD signal in the 210–222 nm region becomes less

negative, approaching zero as the random-coil contribution dominates. The magnitude of this change directly reports the extent of secondary-structure loss.

3. METHODS

3.1 Sample Preparation

Four solutions were prepared as in the fluorescence experiment, following the course manual [5]. Each BSA solution contained 25 µL of 12 mg/mL BSA in 2975 µL of solvent, and each lysozyme solution 100 µL of 3 mg/mL lysozyme in 2900 µL. The solvent was 1×PBS for the native samples and 0.1 M HCl/KCl at pH 1 for the acid samples.

All solutions had final protein concentrations of approximately 0.1 mg/mL.

3.2 Instrument and Scan Parameters

CD spectra were recorded on a JASCO J-810 spectropolarimeter (Rev. 1.00) [6]. The nitrogen purge was set to ≈ 10 psi with a flow rate of 3 L/min, and the xenon lamp was allowed 30 minutes to stabilise before data collection. The high-tension (HT) voltage was checked to be in the 200–300 V range before scanning. The scan parameters were:

TABLE I. JASCO J-810 scan parameters for CD measurements.

Parameter	Value
Sensitivity	100 mdeg
Scan range	250 – 200 nm
Data pitch	1 nm
Scanning speed	100 nm/min (continuous)
Response time	1 s
Bandwidth	1 nm
Accumulations	5
Delay time	0 s
Path length	1 cm (quartz cuvette)
Minimum sample volume	3 mL

Five-scan accumulation averages out random noise while keeping the total measurement time manageable. The 1 nm bandwidth balances spectral resolution against signal-to-noise for the broad protein CD bands.

The four samples are labelled sol1 (BSA, native), sol2 (BSA + acid), sol3 (lysozyme, native) and sol4 (lysozyme + acid) throughout.

4. RESULTS

Fig. 1 shows the CD spectra for all four samples over the 200–250 nm range. The BSA spectra (sol1, sol2) have the strongest negative ellipticity, consistent with BSA's high α-helical content. Both lysozyme spectra (sol3, sol4) show weaker negative ellipticity, reflecting lysozyme's lower α-helix fraction.

Fig. 2 shows the raw JASCO Spectra Analysis output, with the CD signal (upper panel) and the high-tension (HT) voltage (lower panel) for all four samples. The HT voltage rises as more light is absorbed by the sample and buffer. At 210 nm it was 487, 528, 589 and 589 V for sol1–sol4, rising to about 700 V at 200 nm. A higher HT means less transmitted light and a noisier

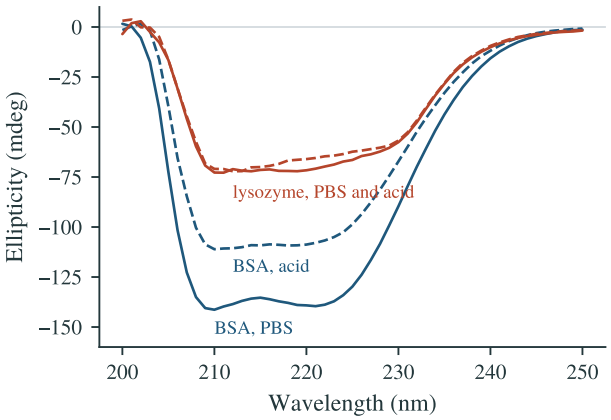


FIG. 1. CD spectra for BSA and lysozyme in native (PBS) and acidified (pH 1) conditions over 200–250 nm, drawn with the same colour and line-style convention as the fluorescence report (BSA blue, lysozyme rust; solid native, dashed acid). The BSA native spectrum (sol1) shows the deepest negative ellipticity. Below ≈ 205 nm, the rising HT voltage indicates strong solvent absorption and reduced data reliability.

signal, so the analysis uses the 210–222 nm band and treats the region below about 205 nm as unreliable. The noise, estimated from the scatter of each spectrum about a smooth local curve, is 0.2–0.4 mdeg.

The ellipticity minima in the 210–222 nm region and the values at the 222 nm helix band are summarised in Table II. Reporting both matters: a comparison at the minimum alone can hide a change in the shape of the band.

TABLE II. Ellipticity (mdeg) at the 210–222 nm minimum, its wavelength $\lambda_{\min}$ (nm), and the ellipticity at the 222 nm α-helix band, for BSA and lysozyme (Lys.) in PBS and at pH 1 (sol1–sol4). Δ is the percentage change in magnitude relative to the native sample, with its uncertainty in the last digit in parentheses; it includes spectral noise only, not differences in concentration between paired samples.

Sample	Min	$\lambda_{\min}$	$\Delta_{\min}$	222 nm	Δ_{222}
BSA	−141.4	210	–	−138.8	–
BSA, acid	−111.1	210	−21.4(3)	−106.7	−23.1(3)
Lys.	−72.8	211	–	−69.8	–
Lys., acid	−72.1	212	−1.0(7)	−64.7	−7.3(7)

5. ANALYSIS

5.1 BSA: Substantial Loss of α-Helical Content

The BSA native spectrum (sol1) has a deep minimum of −141.4 mdeg at 210 nm. On acidification (sol2) it weakens to −111.1 mdeg at the same wavelength, a (21.4 ± 0.3)% reduction in magnitude; the 222 nm band falls by a similar (23.1 ± 0.3)%. Both helix bands weaken together, which indicates a substantial loss of α-helix.

At pH 1, protonation of carboxylate side chains (Asp, Glu) and histidine residues removes the salt bridges and weakens the hydrogen bonding that stabilise the helices. The spectrum does not collapse to a random-coil signature, which would be nearly zero between 210 and 220 nm [5], so the acidified protein retains much of its secondary structure. This is consistent with the acid-expanded F and E forms of albumin, which lose part of their helicity without unfolding completely [3].

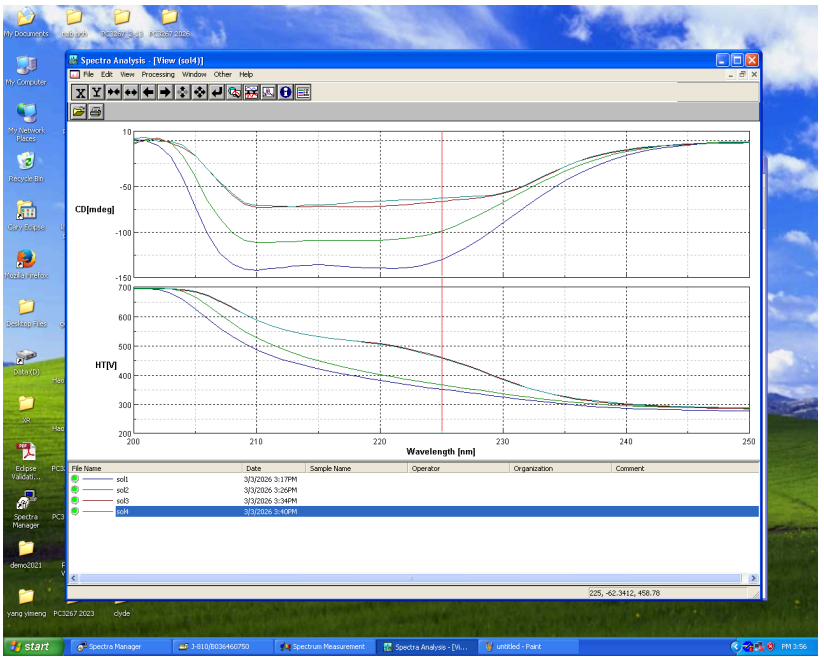


FIG. 2. Raw JASCO J-810 Spectra Analysis output showing CD ellipticity (upper panel) and HT voltage (lower panel) for all four samples. The legend lists the four samples, sol1–sol4.

5.2 Lysozyme: A Smaller but Measurable Change

The lysozyme native spectrum (sol3) has a minimum of -72.8 mdeg at 211 nm. On acidification (sol4) the minimum is -72.1 mdeg at 212 nm, a change of $(1.0 \pm 0.7)\%$, consistent with zero. The 222 nm band, however, weakens from -69.8 to -64.7 mdeg, a $(7.3 \pm 0.7)\%$ reduction. The band therefore changes shape: its deepest point barely moves while its long-wavelength side weakens. Comparing the minima alone would understate the change.

Lysozyme thus loses some helical signal on acidification, but much less than BSA. Its four disulfide bonds cross-link the chain and limit how far it can unfold when protonation weakens its non-covalent interactions [4], which is a plausible reason for the smaller response.

5.3 Comparison

Under the same acid treatment, BSA loses 21–23% of its ellipticity and lysozyme 1–7%, depending on the wavelength compared (Table II). The difference is large compared with the noise, so lysozyme's secondary structure is clearly more resistant to pH 1. The comparison is consistent with stabilisation by disulfide bonds, but it does not isolate them: the two proteins also differ in size, sequence and charge.

At equal mass concentration (0.1 mg/mL), the BSA signal is about twice that of lysozyme, consistent with BSA's larger helical fraction: α-helices give the strongest far-UV CD signal of the common secondary structures [2].

6. DISCUSSION

The CD results give a secondary-structure view that complements the fluorescence spectroscopy, which probes the environment of tryptophan residues. For BSA, both techniques show substantial acid-induced change: fluorescence shows a 19 nm blue shift and a 24% intensity loss, and CD shows a loss of α-helix. BSA at pH 1 therefore undergoes both secondary and tertiary structural change.

For lysozyme, CD shows a modest secondary-structure change, while fluorescence shows a 28% intensity decrease without a wavelength shift. Together these suggest that acid mainly alters the quenching of lysozyme's tryptophan residues, for example through solvent access to a loosened core, while most of its backbone structure remains.

The uncertainties quoted above come from spectral noise alone. Each native/acid pair used the same stock solution, pipette volumes and 1 cm cuvette, so the path length cancels in the percentage changes. Pipetting differences between the paired samples do not cancel, and because no sample was prepared twice they could not be quantified. Five-scan accumulation reduces random noise but not baseline drift.

7. CONCLUSION

Acidification to pH 1 reduces the far-UV ellipticity of BSA by about 21–23%, a substantial loss of α-helix, and that of lysozyme by 1–7%, depending on the wavelength compared. Lysozyme's secondary structure is therefore much more resistant to acid, consistent with the stabilising effect of its four disulfide bonds. The main limitations are the single preparation of each sample and the single comparison of two proteins that differ in more than their disulfide content. Repeat preparations would quantify the concentration uncertainty; measuring lysozyme with its disulfide bonds reduced would test their role directly; and a shorter path-length cell would extend the spectra below 200 nm, where the random-coil band lies, allowing the secondary-structure fractions to be estimated by spectral deconvolution.

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